Calum Buchanan

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Employment

Visiting Assistant Professor, University of Denver (DU), Denver, CO, USA — 2025 – present

Predoctoral Fellow, University of Vermont (UVM), Burlington, VT, USA — 2024 – 2025

Vermont Space Grant Consortium Graduate Fellowship

Graduate Primary Instructor, UVM — 2019 – 2021, 2024

Graduate Research Assistant, UVM — 2022 – 2024

NASA EPSCoR Award

Education

PhD in Mathematics, UVM — May 2025

Dissertation: *Odd Covers and Graph Saturation*

Advisor: Puck Rombach

MSc in Mathematical Sciences, UVM — May 2022

BA in Mathematics and English, UVM — Dec 2017

Research

Fields of interest

Combinatorics; Graph theory

Publications

On cycles in multiset permutations, parking functions, and related structures, with Fabian Burghart, Stephan Wagner, and Mei Yin; in K. Panagiotou (Ed.), *37th International Conference on Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms (AofA 2026)*, volume 381 of Leibniz International Proceedings in Informatics (LIPIcs), pages 16:1-16:17, Dagstuhl, Germany, 2026.

On odd covers of cliques and disjoint unions, with Alexander Clifton, Eric Culver, Péter Frankl, Jiaxi Nie, Kenta Ozeki, Puck Rombach, and Mei Yin; *Journal of Graph Theory*, 2026.

Path odd-covers of graphs, with S. Borgwardt, E. Culver, B. Frederickson, P. Rombach, and Y. Yoo; *Discrete Mathematics*, 439(9):115078, 2026.

Trail Trap: A variant of Partizan Edge Geography, with M. Carr, A. Clifton, S. G. Hartke, V. Iršič Chenoweth, N. Sieger, and R. Whitman; *Discrete Applied Mathematics*, 379:79–95, 2026.

A lower bound on the saturation number and a strengthening for triangle-free graphs, with P. Rombach; *The Electronic Journal of Combinatorics*, 32(3), #P3.14, 2025.

Tropical geometry of Rado matroids, with R. Danner; *Discrete Mathematics Letters*, 14:27–30, 2024.

Node placement to maximize reliability of a communication network with application to satellite swarms, C.B., J. Bagrow, P. Rombach, and H. Ossareh; in *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Honolulu, Oahu, HI, USA; 3466–3473.

Toughness of recursively partitionable graphs, with B. Du Preez, K. E. Perry, and P. Rombach; *Theory and Applications of Graphs*, 10(2), Article 4, 2023.

Odd covers of graphs, with A. Clifton, E. Culver, J. Nie, J. O'Neill, P. Rombach, and M. Yin; *Journal of Graph Theory*, 104:420–439, 2023.

On the last new vertex visited by a random walk in a directed graph, with P. Horn and P. Rombach; *Discrete Mathematics Letters*, 11:96–98, 2023.

Subgraph complementation and minimum rank, with C. Purcell and P. Rombach; *The Electronic Journal of Combinatorics*, 29(1), #P1.38, 2022.

Accepted for publication

On fixing and distinguishing numbers of trees, with P. Dankelmann, I. Harris, P. Horn, K. E. Perry, and E. Rivett-Carnac; to appear in *Discrete Mathematics*. (Preprint: *arXiv.2603.23820*, 2026.)

A Hierarchical Control Architecture for Optimal Guidance and Navigation of Satellite Formations, Y. Pedari, H. Basu, C.B., J. Bagrow, L.D. Espinosa, M.R. Almassalkhi, and H.R. Ossareh; to appear in *Journal of Control, Guidance, and Dynamics*, 2026.

Preprints

Vertex addition to a ball graph with application to reliability and area coverage in autonomous swarms, C.B., P. Rombach, J. Bagrow, and H. R. Ossareh. (Preprint: *arXiv:2506.19197*, 2025.)

Technical reports

One Algorithm to Rule Them All, with N. Brown, N. Chicoine, B. Emerick, A. Emmanuel, C. Hammond, K. Oluwasegun, O. Pomerenk, L. Schneegans, and H. Taboe. *Mathematics in Industry Reports* (submitted April 2025).

Teaching*

Visiting Assistant Professor, DU

- MATH 2060: Elements of Linear Algebra — W 2026
- MATH 1150: Foundations Seminar on the Mathematics of Voting and Counting — W, S 2026
- MATH 1953: Calculus III — F 2025, S 2026
- MATH 1951: Calculus I — F 2026

Graduate Primary Instructor, UVM

- MATH 1212: Fundamentals of Calculus I — F 2019, 2020, 2021
- MATH 1077: Exploring Modern Mathematics — S 2020, 2021, 2024

Instructor of Conversational English — 2018 – 2019

Collèges Le Grand Beauregard and La Coutancière, La Chapelle sur Erdre, France

Teaching Assistant — 2017, 2018

Johns Hopkins Center for Talented Youth, Bristol, RI

Course: Paradoxes and Infinities

Tutor

UVM Tutoring Center, Burlington, VT — 2016 – 2018

*DU's Fall, Winter, and Spring quarters denoted F, W, and S, respectively; UVM's Fall and Spring semesters similarly.

Math Help Sessions, UVM 2019 – 2024

Mentoring

Graduate Mentor, Polymath Jr. REU 2024

Project group leader, Graduate Research Workshop in Combinatorics 2021, 2022

Awards

Vermont Space Grant Consortium Graduate Fellowship, UVM 2024 – 2025

Nam Sang Kil Scholarship in Mathematics, UVM 2023

Graduate Teaching Assistant of the Year, Lecture Instruction Category, UVM 2021

John F. Kenney Award, Excellence in mathematics, UVM 2021

AMS Joint Mathematics Meetings Graduate Student Travel Grant 2021, 2024

AMS Graduate Student Sectional Travel Grant 2023

UVM Graduate College Mini-Grant 2022, 2023

Nominated to attend Mathematical Sciences Research Institute (MSRI), Berkeley, CA 2022
Random Graphs summer graduate school

The Phi Beta Kappa Society, Alpha of Vermont 2017

Service

Math Outreach

Lectured at Governor's Institutes of Vermont and MathCounts NW VT Chapter Competition 2023
Prepared lessons on combinatorics for high school students (materials available on website)

Graduate Student Council 2022 – 2024
College of Engineering and Mathematical Sciences, UVM

Organizing

DU Graduate Colloquium, Denver, CO (organizer) 2025 – 2026

DU Analysis and Dynamics Seminar, Denver, CO (co-organizer) 2025 – 2026

Joint Mathematics Meetings, Seattle, WA 2025
AMS Special Session on Graphs, Hypergraphs, and Extremal Combinatorics (co-organizer)

AMS Spring Central Sectional Meeting, Cincinnati, OH 2023
Special Session on Extremal Graph Theory (co-organizer)

UVM Combinatorics Seminar, Burlington, VT (co-organizer) 2023

Workshops

SAMSA Masamu Advanced Study Institute 2020, 2021, 2023, 2025, 2026

15th Emléktábla Workshop – Extremal combinatorics group 2024

SIAM Mathematical Problems in Industry Workshop	2024
Graduate Research Workshop in Combinatorics	2021, 2022
MSRI Random Graphs Summer Graduate School	2022

Talks Given

34th Cumberland Conference, Auburn University, Auburn, AL — May 2026
Masamu Virtual CRN Colloquium — Apr 2026
AMS Spring Eastern Sectional Meeting,* Boston College, Boston, MA — Mar 2026
DU Analysis and Dynamics Seminar,* Denver, CO — Oct 2025
AMS Spring Eastern Sectional Meeting,* Hartford, CT — Apr 2025
Iowa State University Discrete Math Seminar,* virtual — Feb 2025
Joint Mathematics Meetings,* Seattle, WA — Jan 2025
AMS Special Session on *Coloring Problems in Combinatorics*
AMS Special Session on *Random Walks on Graphs and Related Parameters*
Summit280, Budapest, Hungary — Jul 2024
NY Graph Theory Workshop,* CUNY Graduate Center, New York, NY — May 2024
NY Combinatorics Seminar,* CUNY Graduate Center, New York, NY — May 2024
AMS Spring Eastern Sectional Meeting,* Howard University, Washington, DC — Apr 2024
AMS Special Session on *Recent Trends in Graph Theory*
Dartmouth-UVM Math Day, Dartmouth College, Hanover, NH — Feb 2024
Joint Mathematics Meetings,* San Francisco, CA — Jan 2024
AMS Special Session on *Extremal and Probabilistic Combinatorics*
Southern Africa Mathematical Sciences Association (SAMSA) Annual Conference — Nov 2023
University of Pretoria, Pretoria, South Africa
IEEE Int. Conf. on Systems, Man, and Cybernetics, Honolulu, HI — Oct 2023
AMS Fall Eastern Sectional Meeting,* University at Buffalo (SUNY), Buffalo, NY — Sep 2023
Governor's Institute of Vermont (outreach),* UVM, Burlington, VT — Jul 2023
Rutgers Graduate Combinatorics Seminar,* virtual — Apr 2023
54th Southeastern International Conference on Combinatorics, Graph Theory & Computing (SEICCGTC), Florida Atlantic University, Boca Raton, FL — Mar 2023
MathCounts Northwest VT Chapter Competition (outreach),* UVM, Burlington, VT — Feb 2023
Iowa State University Discrete Math Seminar,* virtual — Jan 2023
Joint Mathematics Meetings, Boston, MA — Jan 2023
AMS Special Session on *Research from the Graduate Research Workshop in Combinatorics (GRWC)*
UVM Combinatorics Seminar,* UVM, Burlington, VT — Nov 2022
Dartmouth Combinatorics Seminar,* Dartmouth College, Hanover, NH — Nov 2022
Virginia Commonwealth University Discrete Math Seminar,* virtual — Oct 2022
SIAM DM22, Carnegie Mellon University, Pittsburgh, PA — Jun 2022
53rd SEICCGTC, virtual — Mar 2022
UMass Amherst Discrete Math Seminar,* Amherst, MA — Mar 2022
UVM Combinatorics Seminar,* UVM, Burlington, VT — Jan 2022

*Indicates an invited talk

AMS Fall Central Sectional Meeting, virtual	Oct 2021
UVM Combinatorics Seminar,* UVM, Burlington, VT	Sep 2021
SIAM DM21, virtual	Jul 2021
Northeast Combinatorics Network Discrete Math Day, virtual	Apr 2021
Graduate Students Combinatorics Conference, virtual	Apr 2021
63rd Midwestern Graph Theory Conference (MIGHTY), virtual	Apr 2021
52nd SEICCGTC, virtual	Mar 2021

Relevant Skills

Coding: Python, LaTeX and TikZ, HTML, SageMath

Languages: English (native), French (advanced)